

SIT292 Module 4 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	1.7/2	
Question 2	1/1	
Question 3	1/1	
Question 4	2/2	
Question 5	1/1	
Question 6	1/1	
Question 7	2/2	
Total	9.7/10	(96%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 4 Self-Assessment
Session ID:	8158182370939113
Exam Start:	Mon Aug 04 2025 14:09:05
Exam Stop:	Mon Aug 04 2025 15:18:33
Time Spent:	1:09:28

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Question 1

Consider the matrix

$$A = \begin{pmatrix} 5 & -3 & -3 \\ -3 & 2 & -3 \\ 5 & -3 & 1 \end{pmatrix}$$

Evaluate the determinant by expanding along row 3.

$$\begin{aligned}
 \det(A) &= \boxed{5} \checkmark \quad \text{Expected answer: } \underline{5} \times \det \\
 &\left(\begin{array}{cc} \boxed{-3} & \boxed{-3} \\ \boxed{2} & \boxed{-3} \end{array} \right) \checkmark \quad \text{Expected answer: } \left(\begin{array}{cc} \underline{-3} & \underline{-3} \\ \underline{2} & \underline{-3} \end{array} \right) + \\
 &\boxed{-3} \times \quad \text{Expected answer: } \underline{3} \times \det \\
 &\left(\begin{array}{cc} \boxed{5} & \boxed{-3} \\ \boxed{-3} & \boxed{-3} \end{array} \right) \checkmark \quad \text{Expected answer: } \left(\begin{array}{cc} \underline{5} & \underline{-3} \\ \underline{-3} & \underline{-3} \end{array} \right) + \\
 &\boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \times \det \\
 &\left(\begin{array}{cc} \boxed{5} & \boxed{-3} \\ \boxed{-3} & \boxed{2} \end{array} \right) \checkmark \quad \text{Expected answer: } \left(\begin{array}{cc} \underline{5} & \underline{-3} \\ \underline{-3} & \underline{2} \end{array} \right) \\
 &= \boxed{5} \checkmark \quad \text{Expected answer: } \underline{5} \times \\
 &\boxed{15} \checkmark \quad \text{Expected answer: } \underline{15} + \\
 &\boxed{-3} \times \quad \text{Expected answer: } \underline{3} \times \boxed{-24} \checkmark \quad \text{Expected answer: } \underline{-24} \\
 &+ \boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \times \\
 &\boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \\
 &= \boxed{4} \checkmark \quad \text{Expected answer: } \underline{4}
 \end{aligned}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and include its determinant in the second line of working.

Score: 1.7/2 ✓**Gap 0**

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 2

- ✗ Your answer is incorrect.

Gap 3

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 4

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 5

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 6

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 7

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 8

- ✗ Your answer is incorrect.

Gap 9

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 10

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 11

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 12

✓ Your answer is correct. You were awarded **0.2** marks.

You scored **1.7** marks for this part.

Question 2

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = -8$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{11} & a_{14} \\ a_{21} & a_{22} & a_{21} & a_{24} \\ a_{31} & a_{32} & a_{31} & a_{34} \\ a_{41} & a_{42} & a_{41} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros
- ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros
- ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

0



Expected answer: 0

Score: 1/1

Gap 0

You chose a correct answer. You were awarded **0.3** marks.

Gap 1

Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 3

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = -15$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} & a_{12} & -5a_{13} & a_{14} \\ a_{21} & a_{22} & -5a_{23} & a_{24} \\ a_{31} & a_{32} & -5a_{33} & a_{34} \\ a_{41} & a_{42} & -5a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☒ 3. Row/Column multiplied by a constant
-
- ☐ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☒ 3. Row/Column multiplied by a constant
- ☐ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

75



Expected answer: 75

Score: 1/1

Gap 0

- ✓ You chose a correct answer. You were awarded **0.3** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 4

For the following matrix,

$$A = \begin{pmatrix} 9 & 8 & 9 \\ 3 & 3 & -3 \\ 7 & 5 & -7 \end{pmatrix}$$

First, calculate the cofactors of A ,

$c_{11} =$ -6 ✓ Expected answer: -6	$c_{12} =$ 0 ✓ Expected answer: 0	$c_{13} =$ -6 ✓ Expected answer: -6
$c_{21} =$ 101 ✓ Expected answer: 101	$c_{22} =$ -126 ✓ Expected answer: -126	$c_{23} =$ 11 ✓ Expected answer: 11
$c_{31} =$ -51 ✓ Expected answer: -51	$c_{32} =$ 54 ✓ Expected answer: 54	$c_{33} =$ 3 ✓ Expected answer: 3

Hence give the adjugate of A ,

$\text{adj}A =$

Rows:

3

 Columns:

3

$$\begin{pmatrix} \div style{border: 1px solid blue; padding: 2px; display: inline-block;">-6} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">101} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">-51} \\ \div style{border: 1px solid blue; padding: 2px; display: inline-block;">0} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">-126} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">54} \\ \div style{border: 1px solid blue; padding: 2px; display: inline-block;">-6} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">11} & \div style{border: 1px solid blue; padding: 2px; display: inline-block;">3} \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">-6} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">101} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">-51} \\ \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">0} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">-126} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">54} \\ \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">-6} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">11} & \div style{border-top: 1px solid black; margin-top: 5px; text-align: center;">3} \end{pmatrix}$$

Score: 2/2 ✓

Gap 0

✓ Your answer is correct. You were awarded **0.15** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 2

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 3

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 4

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 5

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 6

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 7

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 8

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 9

- ✓ Your answer is correct. You were awarded **0.65** marks.

You scored **2** marks for this part.

Question 5

Use determinants to find a value of a for which the matrix is **not** invertible.

$$\begin{pmatrix} 1 & -3 & 7 \\ 3 & -3 & 6 \\ 5 & 9 & a \end{pmatrix}$$

Enter the your value of a exactly, using fractions if required.

$$a = \boxed{-25} \checkmark$$

Expected answer: -25

Score: 1/1

Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Question 6

Let $A, B \in \mathbf{M}_{3,3}(\mathbb{R})$. Suppose that $\det(A) = 4$ and $\det(B) = -2$.

$$\text{i) } \det(AB) = \boxed{-8} \checkmark$$

Expected answer: -8

$$\text{ii) } \det(2A) = \boxed{32} \checkmark$$

Expected answer: 32

$$\text{iii) } \det(A^T B^{-1}) = \boxed{-2} \checkmark$$

Expected answer: -2

Score: 1/1

Gap 0

Your answer is correct. You were awarded 0.3 marks.

Gap 1

Your answer is correct. You were awarded 0.35 marks.

Gap 2

Your answer is correct. You were awarded 0.35 marks.

You scored 1 marks for this part.

Question 7

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 3 & -3 & -6 & 9 \\ 3 & -4 & -8 & 6 \\ -2 & 3 & 6 & -3 \end{pmatrix}$$

Give the matrices L and U that result.

$L =$

Rows: Columns:

$$\begin{pmatrix} \boxed{3} & \boxed{0} & \boxed{0} \\ \boxed{3} & \boxed{-1} & \boxed{0} \\ \boxed{-2} & \boxed{1} & \boxed{1} \end{pmatrix} \quad \checkmark$$

Expected answer: $\begin{pmatrix} \underline{3} & \underline{0} & \underline{0} \\ \underline{3} & \underline{-1} & \underline{0} \\ \underline{-2} & \underline{1} & \underline{1} \end{pmatrix}$

U =

Rows: 3 Columns: 4

$$\begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \checkmark$$

Expected answer:

$$\begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Score: 2/2 **Gap 0** Your answer is correct. You were awarded 1 mark.**Gap 1** Your answer is correct. You were awarded 1 mark.You scored **2** marks for this part.

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